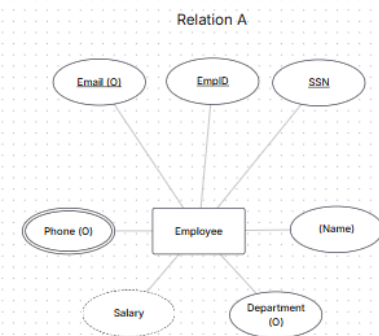


Lab 2 week – Amangeldina Sabina

Task 1

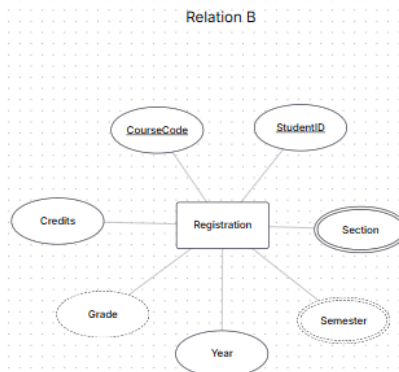
task 1



Primary key: EmpID

Super key: EmpID, SSN, Email, Phone, Name, Department, Salary

Candidate key: EmpID, SSN, Email

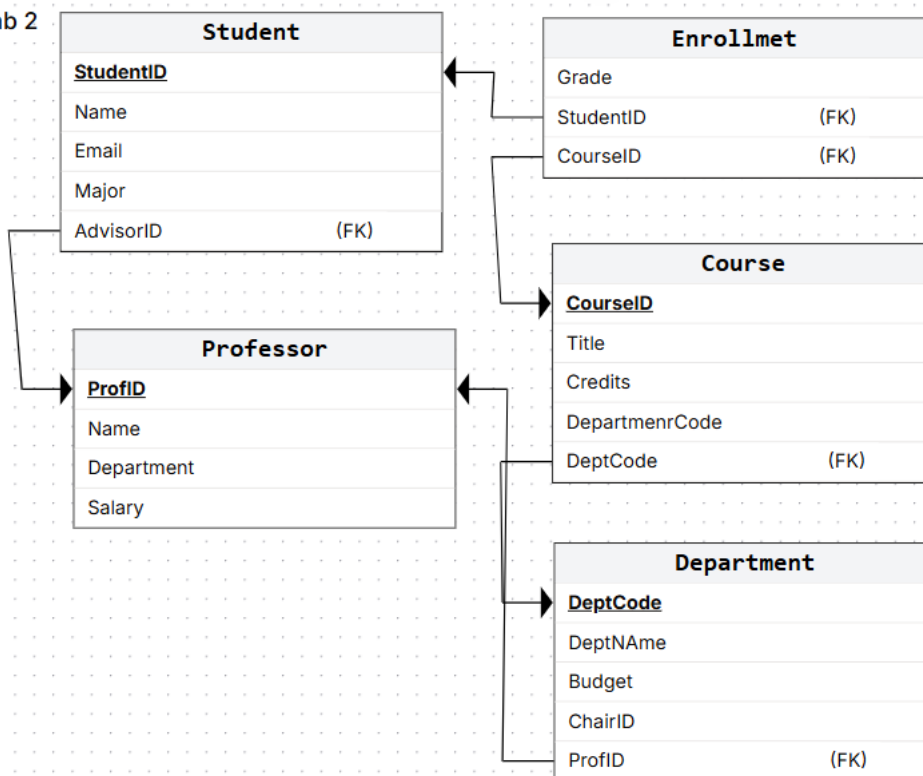


Primary key: StudentID / Super key: StudentID, CourseCode, Section, Semester, Year

Candidatekey: StudentID, CourseCode

Task 1.2

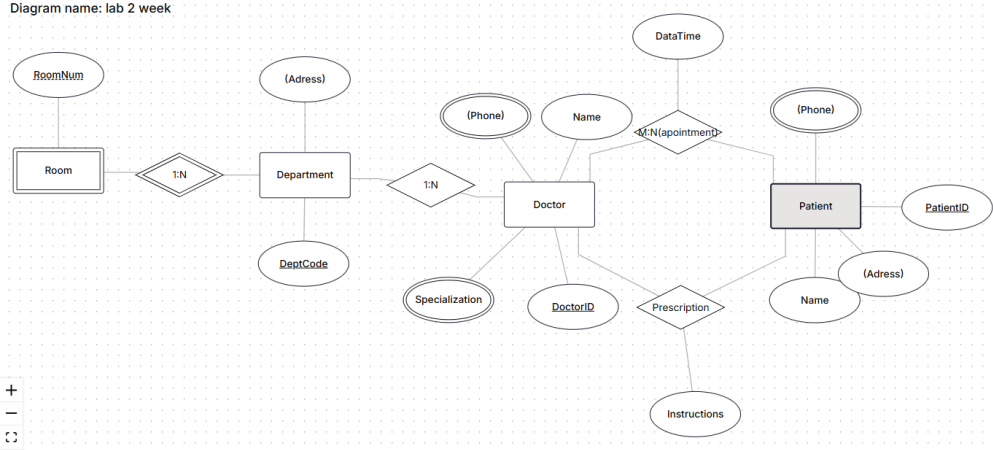
lab 2



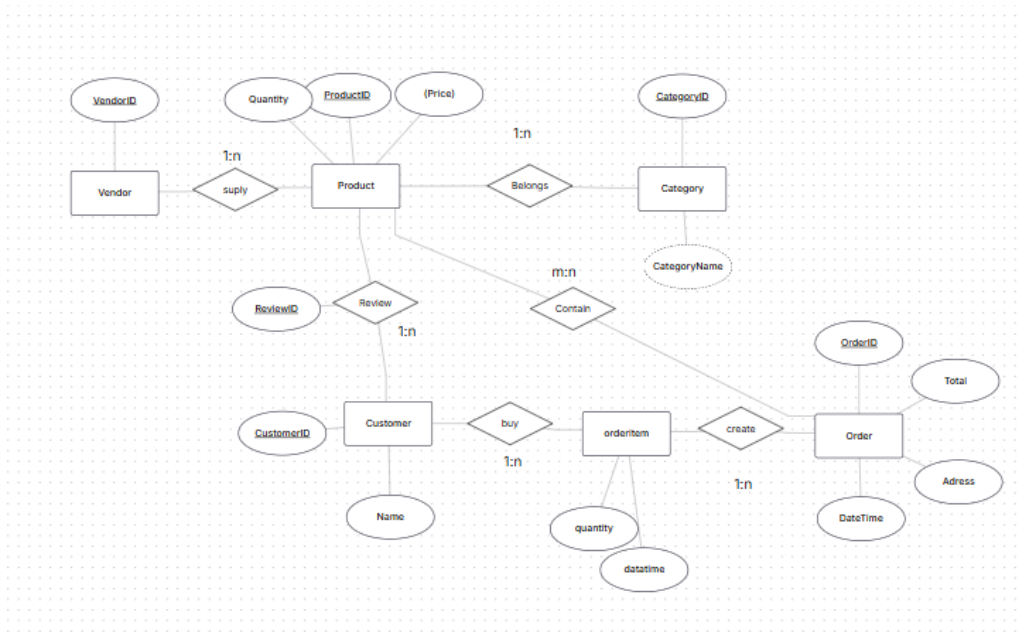
Part 2

Task 2.1

Diagram name: lab 2 week



Task 2.2



Task 4

- StudentProject(StudentID, StudentName, StudentMajor, ProjectID, ProjectTitle, ProjectType, SupervisorID, SupervisorName, SupervisorDept, Role, HoursWorked, StartDate, EndDate)

Identify functional dependencies

- StudentID → StudentName, StudentMajor
- ProjectID → ProjectTitle, ProjectType, SupervisorID
- SupervisorID → SupervisorName, SupervisorDept
- {StudentID, ProjectID} → Role, HoursWorked, StartDate, EndDate

Identify problems

- Student information (Name, Major) repeats for every project. Project information repeats for every student
- Supervisor information repeats every project uses that supervisor. Update the supervisor's department will cause data inconsistency
- No opportunity to add new student, due to StudentID and ProjectID, forms primary key.

1nf

All attributes contain single, atomic values, and there are no repeating groups.

StudentProject(StudentID, ProjectID, StudentName, StudentMajor, ProjectTitle, ProjectType, SupervisorID, SupervisorName, SupervisorDept, Role, HoursWorked, StartDate, EndDate)

2nf

Primary Key: {StudentID, ProjectID}

Student(StudentID, StudentName, StudentMajor)

Project(ProjectID, ProjectTitle, ProjectType, SupervisorID, SupervisorName, SupervisorDept)

StudentProject(StudentID, ProjectID, Role, HoursWorked, StartDate, EndDate)

3nf

StudentID (PK), StudentName, StudentMajor

SupervisorID (PK), SupervisorName, SupervisorDept

ProjectID (PK), ProjectTitle, ProjectType, SupervisorID (FK -> Supervisor.SupervisorID)

StudentID (PK, FK -> Student.StudentID)

ProjectID (PK, FK -> Project.ProjectID)

Role, HoursWorked, StartDate, EndDate

Transitive Dependencies:

- SupervisorID -> SupervisorName, SupervisorDept

Task 4.2

1.

Primary Key: StudentID, CourseID

StudentID alone only determines StudentMajor.

CourseID represents a course section; according to the business rules, a section determines InstructorID, TimeSlot, and Room.

Room determines Building.

A student can enroll in multiple courses, so CourseID alone cannot identify a row. Together, {StudentID, CourseID} uniquely identifies a student's enrollment in a specific course section and all associated details.

2.

StudentID->StudentMajor

CourseID->CourseName,InstructionID,TimeSlot,Room

InstructionID->InstructionName

Room->Building

3. A relation (table) is in BCNF if and only if, for every non-trivial functional dependency $X \rightarrow Y$, the left-hand side (X) is a superkey of the table. This table is not BCNF.

StudentMajor is depend on StudentID and CourseID

Room determinated by Building

InstructorName determinates InstructorID

4.

Student (StudentID, StudentMajor)

Instructor (InstructorID, InstructorName)

Room (Room, Building)

Course (CourseID, CourseName, TimeSlot, InstructorID, Room)

Enrollment (StudentID, CourseID)

Task 5

